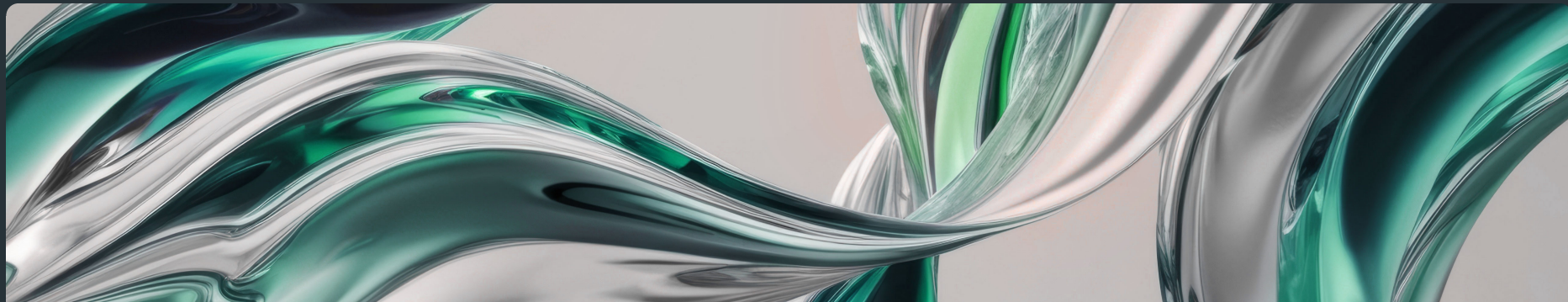


March 2025



Leveraging the Open Science Operating System to build virtual environments for science and education

Sandeep Chandra, Stack Science, San Diego Supercomputer Center



01 Open Science Operating System

02 Use Cases

03 Services & Engagement

04 SGX3 & NAIRR Pilot Engagement

05 Q&A

Introducing the

Open Science Operating System

A unified, end-to-end research cyberinfrastructure solution.

Accelerate your scientific impact.

Free up more time for discovery and innovation by letting experts handle the technical details.

Build collaborative academic communities.

Make it easy for researchers and students alike to find, share, and build upon research.

Ensure the long-term continuity of your research.

Secure ongoing grant funding and keep critical data securely in one place for years to come.

Introducing the

Open Science Operating System

A unified, end-to-end research cyberinfrastructure solution.

Core Capabilities



Advisement

Advise on research cyberinfrastructure, compliance, and sustainability to launch your project and build academic communities.



Science Gateways

Interactive web-based science environments for storing content and datasets, accessing compute, and running applications.



Custom Development

Comprehensive development and design consulting where our developers are embedded in your research projects.

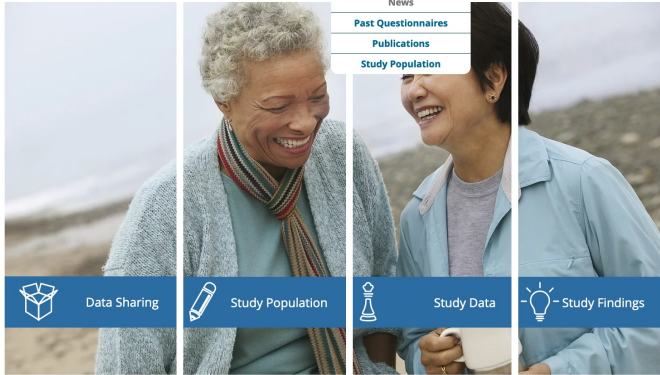


Managed Cyberinfrastructure

Specialized services designed to meet stringent mandates and eliminate compliance uncertainty.

Search...

- CTS Data
- Frequently Asked Questions
- Meet the Team
- News
- Past Questionnaires
- Publications
- Study Population



Search this website ... Search



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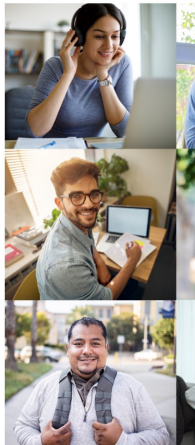
GET STARTED

You have the ability to opt-in to be
invited to participate in one of our
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ude 1) establishment of a
or scanner platforms; 2) central
of data; 3) rigorous quality control
processing and information
integration of longitudinal
aring of data and tools.

edicated to an open approach to
ssible to all levels of inquiry,
methodology driven software
t document ongoing data collection,
n developed following the ABCD
practices developed by the ABCD

the following links for Imaging Analysis

Contact Us

Principal Investigator: Dr.
Anders Dale
**Associate Director,
Bioinformatics:** Dr.
Rongguang Yang

Email:
roy002@health.ucsd.edu

Location

**University of California, San
Diego**
9500 Gilman Drive
La Jolla, CA 92093



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The Continuous Manufacturing Knowledge Center is the go-to place to learn about continuous manufacturing. Whether you have advanced knowledge of continuous manufacturing or are just getting started, the CMKC is the place to connect with experts in the industry, find the latest research, and deepen your understanding of this powerful technology. Gain access to:

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INSPIRING RESEARCH FOR HEALTH EQUITY AND WELL-BEING

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Search, Visualize, Analyze, Download

An open access data, tools, and compute resource operating on NeuroElectroMagnetic data made available in OpenNeuro.org.

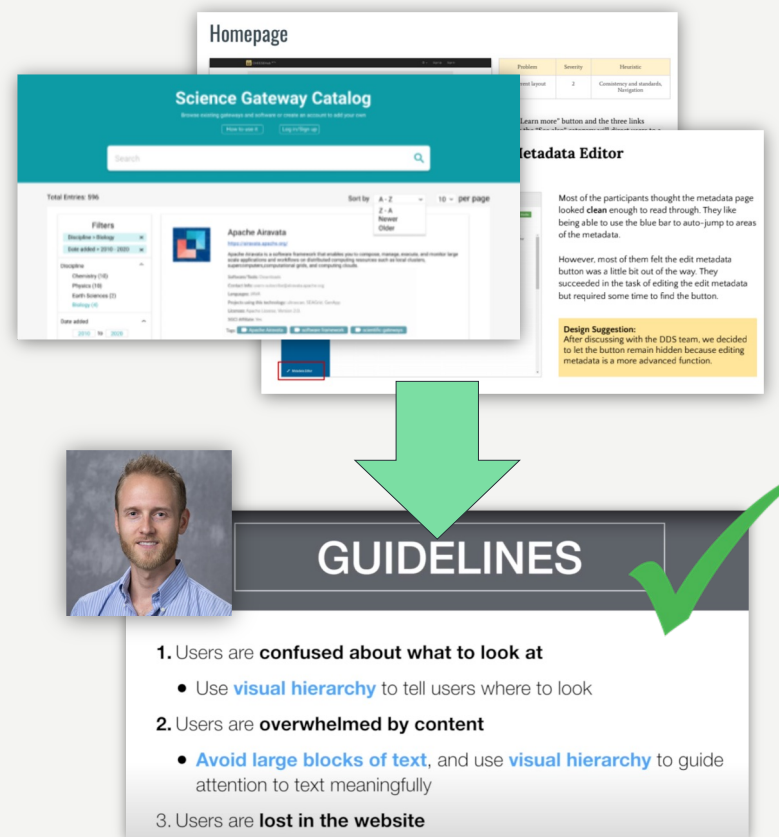
Search for Datasets and Tools

Search

| SGX3 Advisement - Designing for Users

- Teams work with Dr. Paul Parsons (Usability Lead at Purdue University) and the consulting team on reviewing user interfaces or prototyping new designs
- Team provides guidelines and outcomes in reports to advise the client

70+ completed engagements



| University of California Risk Data Management System (RDMS)

UNIVERSITY OF CALIFORNIA Office of the President

CHALLENGE

UC Office of the President business recognized the need for single, centralized risk management system. The IT organization was not prepared for the scale and complexity of managing different data types with different levels of compliance requirements for the entirety of the UC system.

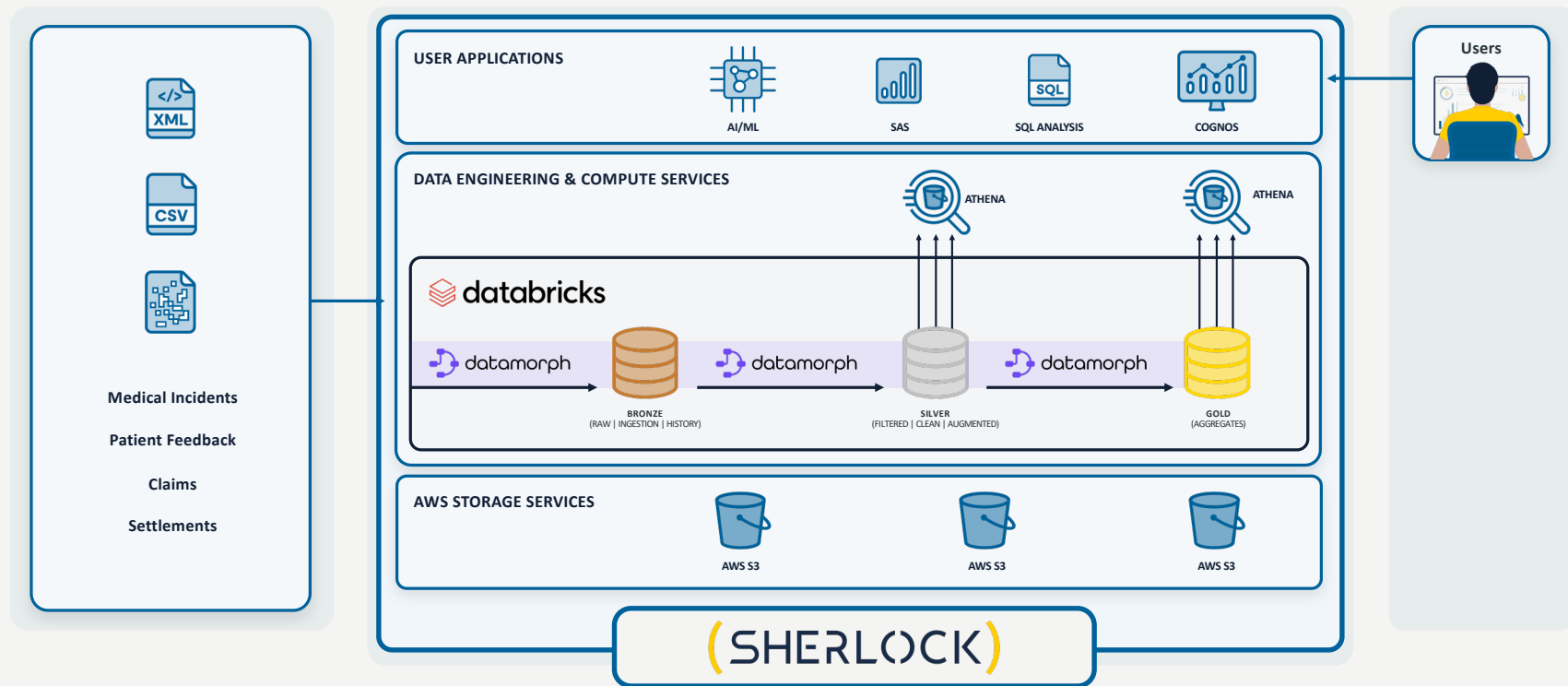
IMPACT: DERISKED UC INVESTMENTS AND BOOSTED CONFIDENCE FOR UNIVERSITY STAKEHOLDERS

Together UC IT & Sherlock provided stakeholders an easy and secure way to access data and reports needed by the Office of the President to confidently make informed decisions, and provide insight for University of California stakeholders, including the California governing bodies.

“The Sherlock team was able to seamlessly take our data and application infrastructure from an open environment to their secure environment in 6 months. This aggressive timeline would not have been feasible absent the expertise and necessary skillset possessed by Sherlock. The end-product has greatly helped secure our data and applications, and we are extremely pleased with the outcome.”

Nilofeur Samuel | Director Risk IT Services, UC Office of the President

University of California Risk Data Management System (RDMS)



| City of Hope – California Teachers Study (CTS)

CHALLENGE

City of Hope was unable to collaborate and effectively get value from research data, due to dated infrastructure, siloed data management approaches, and security and compliance concerns across large-scale cancer epidemiology studies.

IMPACT: INCREASED RESEARCH VELOCITY AND DATA REUSE WITH A MODERN DATA MANAGEMENT SOLUTION

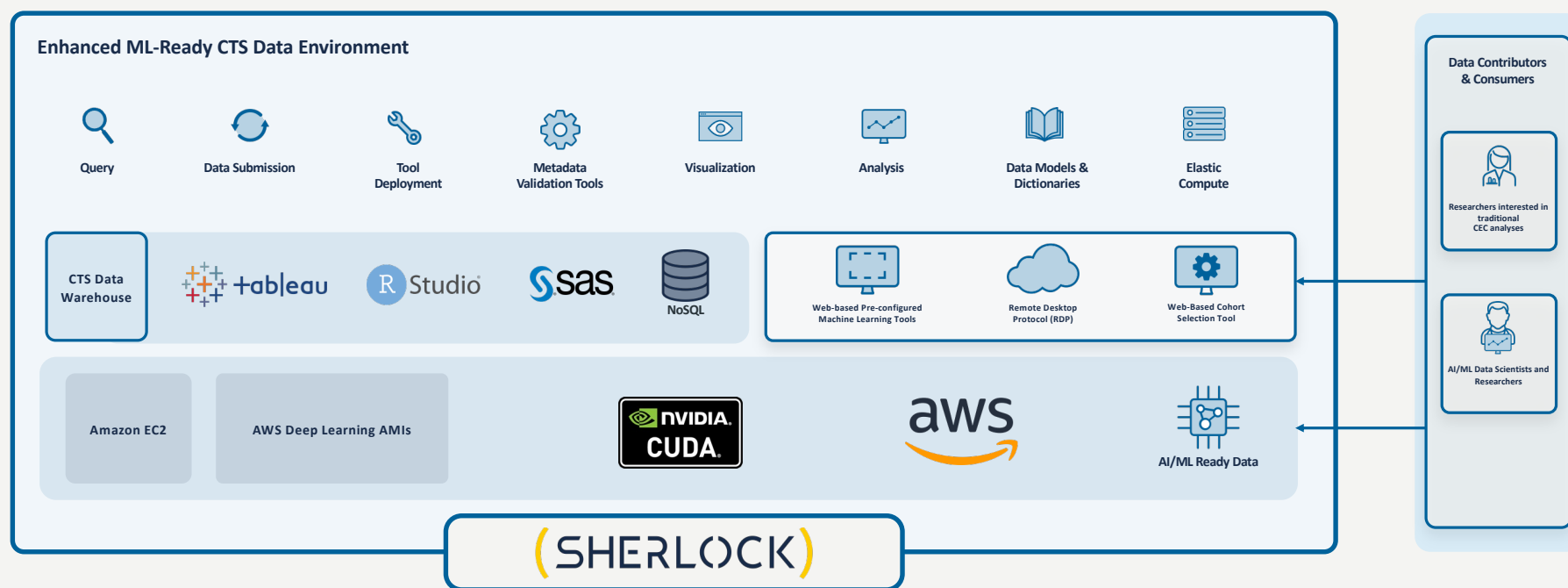
City of Hope and Sherlock built an evergreen data product that transformed data collection, storage, and reusability for high- impact research. The project increased security and efficiency while also reducing ongoing research costs.



“Sherlock Cloud is entirely HIPAA- and FISMA-compliant, meaning the Sherlock team has specific experience providing the type of data security that our CTS Data Mart and research require. Our partnership with Sherlock will modernize the long-standing CTS project through Sherlock’s innovative data management solutions, which will make the California Teachers Study an example for other studies like this.”

***James Lacey | Director of Cancer Etiology,
Beckman Research Institute, City of Hope***

City of Hope – California Teachers Study (CTS)



| Nihon Kohden Digital Health Platform



CHALLENGE

Patient monitoring devices collect a wide variety of data including vital signs, alarms and waveforms. Nihon Kohden product team wanted to develop innovative solutions that could benefit clinicians providing patient care. However, healthcare HIPAA data requirements prevent the Nihon Kohden from being able to quickly and confidently build a platform for securely collecting, storing, and preparing device data for generating insights .

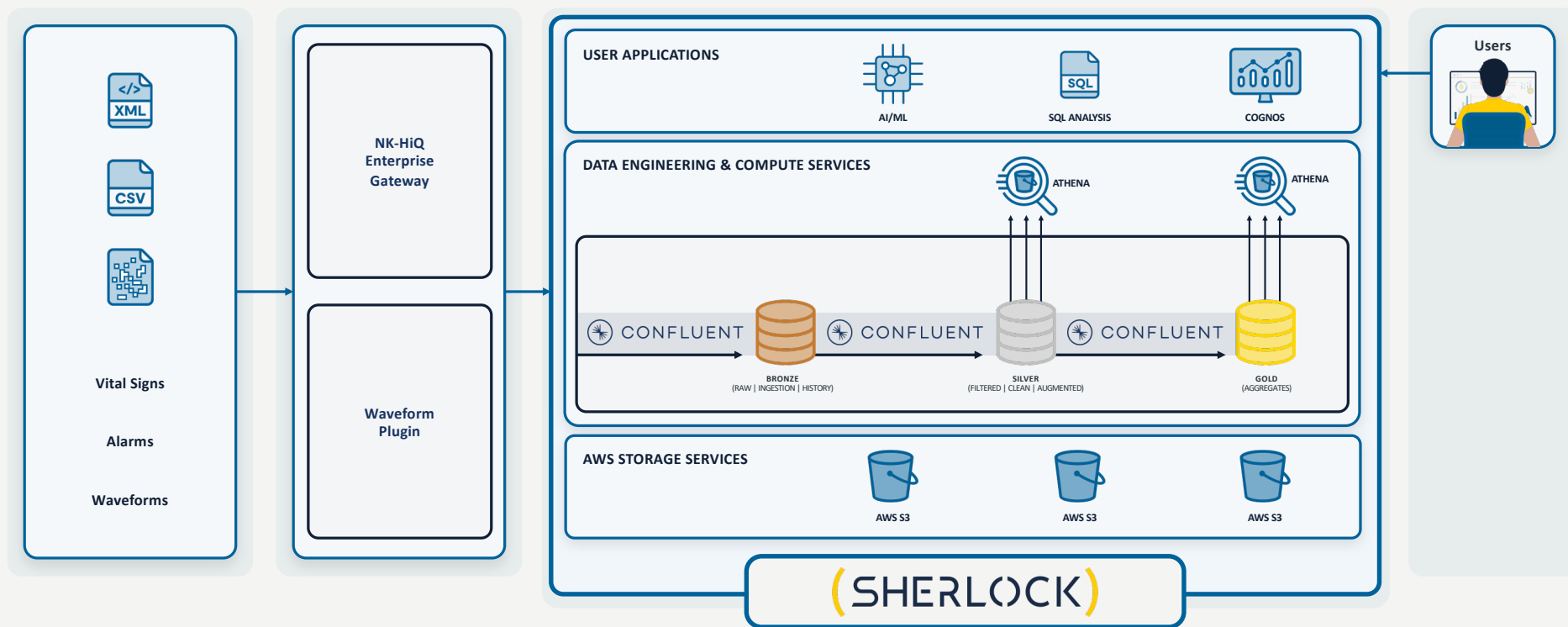
IMPACT: CONFIDENTLY INCREASED INNOVATION AND CUSTOMER TRUST WITH SHERLOCK CLOUD

Nihon Kohden's product team created innovative, and data driven software solutions for the full range of Nihon Kohden's medical devices. Nihon Kohden brought these new solutions to market, confident that they meet the compliance and regulatory needs of their healthcare provider customers by building them on Sherlock's solution.

“As part of the NK technology roadmap, we continuously explore options for managing data securely, more economically and with greater flexibility and scalability. Through our partnership with Sherlock, we were able to leverage the core competencies of each team toward a successful implementation of a modern, cloud-based and secure data lake platform that could serve as a strategic and forward-looking approach to data management and product development.”

Abel Lin | Director, Integration & Data Engineering, Nihon Kohden Digital Health Solutions Inc.

Nihon Kohden Digital Health Platform



Our products & services:



Advisement Core - Center of Excellence to Extend Access, Expand the Community, and Exemplify Good Practices for CI Through Science Gateways.



Custom Development Core - Software development, cyberinfrastructure operations, and product consulting for tailored solutions.



Science Gateways Core - Platform for hosting, sharing, and publishing software tools, data, and fostering scientific collaboration.

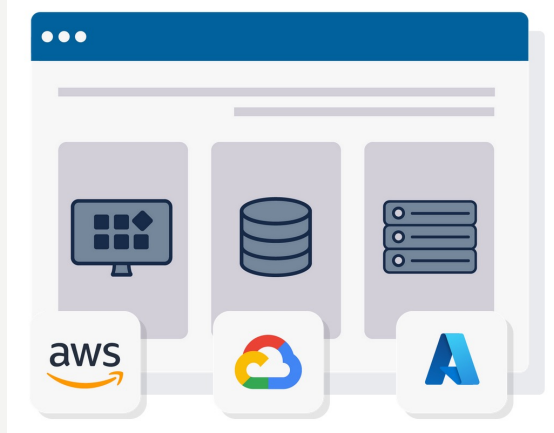


Managed Infrastructure Core - Secure research environments for managing sensitive data, compliant with HIPAA, FISMA, CUI/CMMC, and more.

Meeting research needs with turnkey solutions for regulatory data management

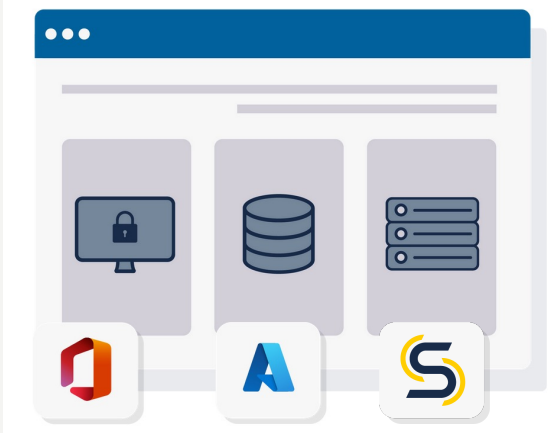
(SHERLOCK) CLOUD SERVICES

- ✓ Compliance standards: HIPAA, FISMA, FERPA
- ✓ Cloud support: AWS, Azure, GCP
- ✓ Use cases: Application hosting, compute, storage, secure desktops, data platforms, custom solutioning



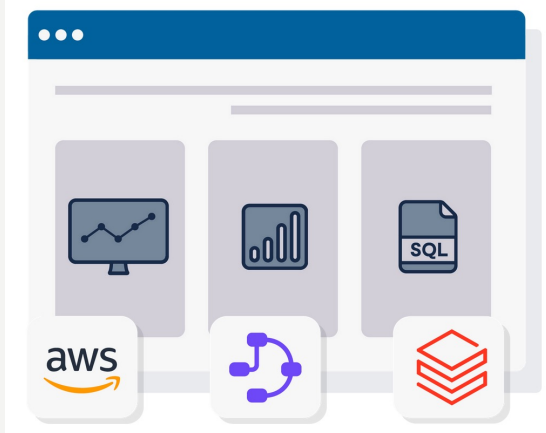
(SHERLOCK) GOVERNMENT SERVICES

- ✓ Compliance standards: CUI, CMMC L2
- ✓ Cloud support: Azure Gov, O365 GCCH
- ✓ Use cases: Compute, storage, secure desktops, productivity apps



(SHERLOCK) DATA MANAGEMENT

- ✓ Compliance standards: HIPAA, FISMA, FERPA
- ✓ Cloud support: AWS
- ✓ Use cases: Platforms for data lakes, data engineering, data science, machine learning



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Gathering
& Solutioning

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Gateways
Deployment

Custom
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Software
Development

Community
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Sustainability

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Maintenance

Regulatory
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Proposal
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UCSF
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External Academia
University of Rhode Island
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Boston University
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Purdue University
Oregon State University
Stony Brook University

Agencies we work with



Thank You

Stack Science

San Diego Supercomputer Center (SDSC)
University of California San Diego (UCSD)
9500 Gilman Drive
Mail Code 0505
La Jolla, CA 92093-0505

info@stackscience.org
stackscience.org